PROBLEM 1.4

Newton's Law of Cooling Modeling

Original statement · ProblemsCh1.pdf, p. 2

Lumped Thermal Model · Variable Identification

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A cup of hot chocolate is initially at temperature $T(0)$ and is left in the room with an ambient temperature $T_A(t)$. The rate of cooling of the hot chocolate is proportional to the difference between the current temperature $T(t)$ and the ambient temperature.

(i) Which are the dependent variable, the independent variable, and the parameters?
(ii) Write a linear differential equation for the temperature of the hot chocolate that takes into account the heat exchange with the room. Such differential equation is known as Newton's law of cooling.

PHASE 2: FUNDAMENTAL FORMULATION

Lumped capacitance thermal balance: Internal thermal energy rate $mc_p\frac{dT}{dt}$ balances convective heat loss $\dot{Q}_{\text{loss}} = hA_s(T - T_A(t))$. Rate of cooling is $-\frac{dT}{dt}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): Identification of Variables and Parameters:

- **Independent variable:** Time t $[s]$ (or $[\text{min}]$, $[\text{h}]$).
- **Dependent variable:** Temperature of the hot chocolate $T = T(t)$ $[^\circ\text{C}]$ or $[\text{K}]$.
- **Parameters:**
  - Cooling constant: $k = \frac{hA_s}{mc_p} > 0$ $[t^{-1}]$.
  - Initial condition: $T(0) = T_0$ $[^\circ\text{C}]$.
  - Ambient temperature: $T_A(t)$ (specified external driving function; acts as a parameter if constant).

Part (ii): Formulation of Newton's Law of Cooling:

The verbal statement dictates that the cooling rate $-\frac{dT}{dt}$ is proportional to $(T(t) - T_A(t))$ with constant $k > 0$:

$$-\frac{dT}{dt} = k(T(t) - T_A(t))$$

Multiplying by -1 and rearranging to standard linear form:

$$\mathbf{\frac{dT}{dt} + kT(t) = kT_A(t)} \qquad (1.4.1)$$

PHASE 4: FINAL RESULT & LINEARITY CHECK

BOXED RESULT

$$\frac{dT}{dt} = -k(T - T_A(t)) \iff \frac{dT}{dt} + kT = kT_A(t)$$

Linearity: In form $a_1(t)T' + a_0(t)T = b(t)$ with $a_1(t) = 1, a_0(t) = k, b(t) = kT_A(t)$. Strictly 1st-order linear ODE.

METHOD OF THE CLASS NOTES [NOTES, PP. 2-3]

The equation is $x' + p(t)x = q(t)$ with $p = k, q = kT_A(t)$, which the three-step algorithm of the notes solves for any T_A. Step 1: $P(t) = \int_0^t k\,d\tau = kt, I(t) = e^{kt}, I(0) = 1$. Step 2: $\int_0^t \left(e^{k\tau}T(\tau)\right)' d\tau = \int_0^t e^{k\tau}kT_A(\tau)\,d\tau$. Step 3: Barrow's rule on the left gives

$$T(t) = T(0)e^{-kt} + e^{-kt}\int_0^t k\,e^{k\tau}\,T_A(\tau)\,d\tau.$$

Problem 1.5 applies it with constant T_A.